

PINE SPRINGS GUADALUPE PASS, TX, USA

WMO: 722620

Lat: 31.831N

Lon: 104.807W

Elev: 5456

StdP: 12.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23055

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.4	23.7	-5.7	5.0	50.5	-1.4	6.2	48.1	50.9	44.4	47.7	42.5	25.1	60	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	23.0	95.3	60.0	92.7	60.2	90.2	60.1	66.8	81.3	65.8	80.5	64.8	79.3	16.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	107.2	68.7	62.2	102.8	68.8	61.0	98.3	69.0	34.5	81.2	33.6	79.9	32.7	78.8	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
47.2	42.3	38.1	14.5	100.5	4.5	1.7	11.2	101.8	8.6	102.8	6.1	103.8	2.8	105.0		
			11.8	68.5	3.3	1.2	9.4	69.4	7.5	70.1	5.6	70.7	3.2	71.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	43.2	47.6	54.0	61.4	69.6	77.8	77.3	76.2	70.6	62.0	51.3	42.7	(6)
	DBStd	14.80	8.11	9.08	9.41	8.31	7.94	5.87	5.00	4.83	6.15	8.06	9.06	8.39	(7)
	HDD50	819	233	140	70	15	2	0	0	0	0	14	92	253	(8)
	HDD65	2995	676	488	350	156	47	1	1	1	23	147	413	692	(9)
	CDD50	4914	23	72	195	359	610	833	846	814	618	386	132	26	(10)
	CDD65	1617	0	1	11	50	191	385	382	350	191	54	2	0	(11)
	CDH74	14401	0	6	91	458	1847	4000	3494	2862	1296	334	13	0	(12)
	CDH80	5818	0	0	8	92	710	1946	1476	1118	411	57	0	0	(13)
(14) Wind	WSAvg	18.9	21.0	21.8	21.3	22.0	20.2	16.8	14.9	14.4	15.7	17.7	20.0	20.7	(14)
(15) Precipitation	PrecAvg	11.9	0.4	0.5	0.4	0.4	0.7	1.1	2.2	2.2	2.1	1.1	0.6	0.6	(15)
	PrecMax	23.0	2.2	1.8	1.3	2.0	3.3	4.8	4.5	5.5	5.4	3.1	2.9	2.8	(16)
	PrecMin	7.3	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	3.9	0.6	0.4	0.3	0.6	0.8	1.0	1.2	1.3	1.7	0.9	0.7	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	74.2	80.5	87.0	94.5	100.4	97.3	95.5	91.7	85.0	75.8	66.6	(19)
		MCWB	44.3	47.6	50.1	54.2	57.0	60.1	61.1	63.1	62.0	59.3	51.5	44.9	(20)
	2%	DB	62.7	69.2	76.6	82.4	90.5	96.7	93.4	91.9	88.0	81.3	71.9	62.6	(21)
		MCWB	43.4	45.7	48.9	51.8	55.8	59.0	61.9	63.6	60.7	56.0	50.0	44.7	(22)
	5%	DB	59.0	65.0	72.7	79.3	87.6	93.6	90.6	89.2	84.4	77.9	67.7	58.5	(23)
		MCWB	41.8	44.2	47.3	50.2	54.8	58.5	62.2	63.0	60.3	54.9	48.3	42.7	(24)
	10%	DB	55.3	61.5	69.0	75.7	84.0	90.6	88.1	86.2	81.6	74.2	63.9	55.0	(25)
		MCWB	40.2	42.8	45.9	49.1	53.5	58.4	61.9	62.6	60.0	53.7	46.9	40.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	50.6	54.4	58.2	62.3	66.3	68.6	68.2	67.4	63.8	56.4	49.9	(27)
		MCDB	58.4	66.7	72.8	72.3	78.6	80.8	82.5	83.6	78.7	74.8	66.6	60.3	(28)
	2%	WB	45.4	47.9	51.2	55.4	60.3	64.7	66.7	66.8	65.7	61.3	52.8	46.3	(29)
		MCDB	57.4	63.4	70.1	72.3	76.6	80.5	81.1	82.6	77.3	71.4	65.0	57.9	(30)
	5%	WB	43.4	45.9	49.1	53.1	58.7	63.6	65.7	65.8	64.3	59.3	50.6	44.1	(31)
		MCDB	56.3	61.4	67.9	71.5	75.3	80.3	80.3	81.6	75.4	69.6	63.2	56.4	(32)
	10%	WB	41.3	44.0	47.1	51.2	56.9	62.3	64.5	64.7	63.1	57.1	48.4	41.9	(33)
		MCDB	54.3	59.2	66.3	70.9	74.1	79.3	79.1	79.9	74.0	68.0	61.3	53.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	20.8	22.2	23.2	22.7	23.0	19.9	19.5	18.9	19.7	19.2	18.2	(35)
		MCDBR	21.8	23.5	25.0	25.5	25.6	25.8	23.0	22.5	22.8	22.2	21.9	21.3	(36)
		MCWBR	11.5	11.3	11.4	11.2	10.3	8.4	7.0	7.0	7.9	9.4	10.1	11.0	(37)
	5% WB	MCDBR	20.3	22.0	23.5	24.4	23.4	22.3	20.4	20.4	19.1	19.2	19.4	19.5	(38)
		MCWBR	11.6	11.2	11.4	11.3	11.0	8.1	7.1	7.3	7.5	9.1	9.9	10.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.234	0.241	0.254	0.280	0.304	0.317	0.345	0.329	0.314	0.271	0.253	0.239	(40)
	taud		2.677	2.647	2.597	2.489	2.465	2.478	2.434	2.486	2.480	2.627	2.609	2.637	(41)
	Ebn at Noon		324	329	329	321	311	305	296	301	303	313	313	316	(42)
	Edh at Noon		23	26	30	34	36	35	36	34	33	26	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1152	1458	1873	2270	2441	2450	2240	2082	1835	1535	1220	1036	(44)
	RadStd		95	118	108	125	121	111	113	93	107	131	100	72	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page